

RAWAL KHIRODKAR

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RESEARCH INTERESTS

Fields: Computer Vision, Machine Learning, Robotics
Topics: Digital Humans, Multimodal Reasoning, Generative AI

EDUCATION

Carnegie Mellon University Ph.D. in Robotics	Aug 2019 – Sep 2023 Advisor: Prof. Kris Kitani
Carnegie Mellon University M.S. in Robotics	Aug 2017 – July 2019 Advisor: Prof. Kris Kitani
Indian Institute of Technology, Bombay Bachelors in Computer Science	Aug 2013 – July 2017 Advisor: Prof. Ramakrishnan

PROFESSIONAL EXPERIENCE

Meta Reality Labs Research Scientist	July 2024 – Present Pittsburgh, PA
Meta Reality Labs Postdoctoral Research Scientist	Sep 2023 – July 2024 Pittsburgh, PA
Meta Reality Labs Research Intern	May 2022 – Aug 2022 Redmond, WA
Amazon Lab126 Research Intern	May 2021 – Aug 2021 Sunnyvale, CA
Amazon Lab126 Research Intern	May 2020 – Aug 2020 Sunnyvale, CA
Trexquant Finance Quantitative Analyst	May 2017 – Aug 2017 Stamford, CT

AWARDS & HONORS

• Best Paper Candidate, ECCV 2024, <i>top 10 papers</i>	2024
• Distinguished Paper Award, CVPR 2023, Egovis Workshop,	2023
• Amazon Graduate Fellowship,	2020
• Government of India Graduate Fellowship, <i>top 25 students</i>	2020–22
• Best Teaching Assistant Honorable Mention, IIT Bombay,	2017
• Indian National Physics Olympiad, <i>top 50 students</i>	2013
• Indian National Maths Olympiad, <i>top 50 students</i>	2013
• NTSE Scholar, Awarded to top 800 amongst 0.5 million students,	2009–13

PUBLICATIONS

- [1] **Sapiens2**
Rawal Khirodkar, He Wen, Julieta Martinez, Yuan Dong, Su Zhaoen, Shunsuke Saito
International Conference on Learning Representations (ICLR), 2026
- [2] **LCA: Large-scale Codec Avatars – The Unreasonable Effectiveness of Large-scale Avatar Pretraining**
Junxuan Li*, **Rawal Khirodkar***, et al.
Conference on Computer Vision and Pattern Recognition (CVPR), 2026
- [3] **DuoMo: Dual Motion Diffusion for World-Space Human Reconstruction**
Yufu Wang, Evonne Ng, Soyong Shin, **Rawal Khirodkar**, Yuan Dong, Su Zhaoen, Jinhyung Park, Kris Kitani, Alexander Richard, Fabian Prada, Michael Zollhöfer
Conference on Computer Vision and Pattern Recognition (CVPR), 2026
- [4] **GenLCA: 3D Diffusion for Full-Body Avatars from In-the-Wild Videos**
Yiqian Wu, **Rawal Khirodkar**, Egor Zakharov, Timur Bagautdinov, Lei Xiao, Su Zhaoen, Shunsuke Saito, Xiaogang Jin, Junxuan Li
arxiv, 2026
- [5] **MHR: Momentum Human Rig**
Meta Reality Labs Research. Mesh representation used by SAM3D-Body.
arxiv, 2025
- [6] **ATLAS: Decoupling Skeletal and Shape Parameters for Expressive Parametric Human Modeling**
Jinhyung Park, Javier Romero, Shunsuke Saito, Fabian Prada, Takaaki Shiratori, Yichen Xu, Federica Bogo, Shoou-I Yu, Kris Kitani, **Rawal Khirodkar**
International Conference on Computer Vision (ICCV), 2025
- [7] **Capture, Canonicalize, Splat: Zero-Shot 3D Gaussian Avatars from Unstructured Phone Images**
Meta Reality Labs Research.
International Conference on Computer Vision (ICCV) Workshop, 2025 (*Best Paper Honorable Mention*)
- [8] **Pippo : High-Resolution Multi-View Humans from a Single Image**
Yash Kant, Ethan Weber, Jin Kyu Kim, **Rawal Khirodkar**, Su Zhaoen, Julieta Martinez, Igor Gilitschen-ski, Shunsuke Saito, Timur Bagautdinov
Conference on Computer Vision and Pattern Recognition (CVPR), 2024 (*Highlight*)
- [9] **Harmony4D: A Video Dataset for In-The-Wild Close Human Interactions**
Rawal Khirodkar, Jyun-Ting Song, Jinkun Cao, Zhengyi Luo, Kris Kitani
Neural Information Processing Systems, NeurIPS (Datasets and Benchmark Track), 2024
- [10] **URAvatar: Universal Relightable Gaussian Codec Avatars**
Junxuan Li, Chen Cao, Gabriel Schwartz, **Rawal Khirodkar**, Christian Richardt, Tomas Simon, Yaser Sheikh, Shunsuke Saito
Special Interest Group on Computer Graphics and Interactive Techniques, SIGGRAPH Asia, 2024
- [11] **Sapiens: Foundation for Human Vision Models**
Rawal Khirodkar, Timur Bagautdinov, Julieta Martinez, Su Zhaoen, Austin James, Peter Selednik, Stuart Anderson, Shunsuke Saito
European Conference on Computer Vision (ECCV), 2024 (*Best Paper Candidate, 5.3k Github Stars*)

- [12] **Ego-Exo4d: Understanding Skilled Human Activity from First-and Third-Person Perspectives**
Ego4D Consortium
Conference on Computer Vision and Pattern Recognition (CVPR), 2024 (Oral Presentation)
- [13] **Real-Time Simulated Avatar from Head-Mounted Sensors**
Zhengyi Luo, Jinkun Cao, **Rawal Khirodkar**, Alexander Winkler, Kris Kitani, Weipeng Xu
Conference on Computer Vision and Pattern Recognition (CVPR), 2024 (Highlight)
- [14] **Dual-Modal 3D Human Pose Estimation using Insole Foot Pressure Sensors**
Erwin Wu, Yichen Peng, **Rawal Khirodkar**, Hideo Koike, Kris Kitani
International Symposium on Mixed and Augmented Reality Adjunct (ISMAR), 2024
- [15] **SolePoser: Full body pose estimation using a single pair of insole sensor**
Erwin Wu, **Rawal Khirodkar**, Hideki Koike, Kris Kitani
ACM Symposium on User Interface Software and Technology, 2024
- [16] **Generalizable Neural Human Renderer**
Mana Masuda, Jinhyung Park, Shun Iwase, **Rawal Khirodkar**, Kris Kitani
Meeting on Image Recognition and Understanding (MIRU), 2024 (Oral Presentation)
- [17] **Multi-Person 3D Pose Estimation from Multi-view Uncalibrated Depth Cameras**
Yu-Jhe Li, Yan Xu, **Rawal Khirodkar**, Jinhyung Park, Kris Kitani
arxiv, 2024
- [18] **EgoHumans: An Egocentric 3D Multi-Human Benchmark**
Rawal Khirodkar, Aayush Bansal, Lingni Ma, Richard Newcombe, Minh Vo, Kris Kitani
International Conference on Computer Vision (ICCV), 2023 (Oral Presentation)
- [19] **Observation-Centric SORT: Rethinking SORT for Robust Multi-Object Tracking**
Jinkun Cao, Xinshuo Weng, **Rawal Khirodkar**, Jiangmiao Pang, Kris Kitani
Conference on Computer Vision and Pattern Recognition (CVPR), 2023
- [20] **Sequential Ensembling for Semantic Segmentation**
Rawal Khirodkar, Brandon Smith, Siddhartha Chandra, Amit Agrawal, Antonio C.
arxiv, 2022
- [21] **Occluded Human Mesh Recovery**
Rawal Khirodkar, Shashank Tripathi, Kris Kitani
Conference on Computer Vision and Pattern Recognition (CVPR), 2022
- [22] **Multi-Instance Pose Networks: Rethinking Top-Down Pose Estimation**
Rawal Khirodkar, Visesh Chari, Amit Agrawal, Ambrish Tyagi
International Conference on Computer Vision (ICCV), 2021
- [23] **RePOSE: Fast 6D Object Pose Refinement via Deep Texture Rendering**
Shun Iwase, Xingyu Liu, **Rawal Khirodkar**, Rio Yokota, Kris Kitani
International Conference on Computer Vision (ICCV), 2021
- [24] **Adversarial Domain Randomization**
Rawal Khirodkar, Kris Kitani
arxiv, 2019

- [25] [Domain Randomization for Scene Specific Object Detection & Pose Estimation](#)
Rawal Khirodkar, Donghyun Yoo, Kris Kitani
Winter Conference on Applications of Computer Vision (WACV), 2019

PATENTS

- [1] [Human Pose Rendering](#)
Shawn Hunt, Kris Makoto Kitani, Jinhyung Park, Shun Iwase, Rawal Khirodkar, Mana Masuda
US Patent App. 18/616,942, 2025
- [2] [Multi-Person 3D Pose Estimation](#)
Jinhyung Park, Yu-Jhe Li, Rawal Khirodkar, Kris Kitani, Shawn Hunt
US Patent App. 18/504,429, 2025

INVITED TALKS

- 3D Human Understanding, *Hosted by Prof. Kristen Grauman, University of Texas at Austin* 2025
- Best Practices to building Foundation Models, *Facebook AI Research* 2024
- Sapiens: Foundation for Human Vision Models, *Meta Reality Labs* 2024
- Building 3D Datasets from Scratch, *Project Aria Workshop, CVPR* 2023
- Egocentric Human Understanding, *Massachusetts Institute of Technology, Graphics Seminar* 2023
- In-the-Wild Human Pose Estimation, *National University of Singapore, Vision Seminar* 2023
- Using Synthetic Data for Long-Tail Problems, *Carnegie Mellon University* 2022

SELECTED MEDIA COVERAGE

SAPIENS

[LearnOpenCV](#), [Hacker News](#), [MarkTechPost](#), [TeqnoVerse](#), [Unite AI](#), [Sentisight News](#)

EGO-EXO4D

[AI-Daily](#), [University of Bristol](#), [BTW-Media](#), [Georgia-Tech](#)

PROFESSIONAL SERVICE

ORGANIZER

- Co-organizer, 3D Human Workshop, *Second Edition, CVPR* 2025
- Co-organizer, 3D Human Workshop, *First Edition, CVPR* 2024

CONFERENCE REVIEWER

NeurIPS, ICML, ICLR, CVPR, ICCV, ECCV, WACV, AACL, SIGGRAPH Asia

JOURNAL REVIEWER

JMLR, IJCV, TPAMI

ADMISSIONS COMMITTEE

Carnegie Mellon University: Masters in Computer Vision (*twice*), Masters in Robotics, Ph.D. in Robotics

RESEARCH MENTORING

- [David Park](#) (CMU RI PhD), 2023–2024
- [Jyun-Ting Song](#) (CMU MSR, now PhD at CMU RI), 2023–2024
- [Jinkun Cao](#) (CMU RI PhD, now at Meta), 2022–2023
- [Shun Iwase](#) (CMU MSR, now PhD at CMU RI), 2020–2021
- [Shengcao Cao](#) (CMU MSR, now PhD at UIUC), 2019–2020
- [Rishi Madhok](#) (CMU MSCV, now at Microsoft), 2018–2019

TEACHING

TEACHING ASSISTANCE

- Computer Vision (16-720), CMU, *Instructor: Srinivasa & Kris* Fall 2021 & 2020, Spring 2018
- Statistical Techniques in Robotics (16-831), CMU, *Instructor: Kris Kitani* Spring 2019
- Math Fundamentals (16-811), CMU, *Instructor: Michael Erdmann* Fall 2019
- Machine Learning (10-601), CMU, *Instructor: Matt Gormley* Fall 2018
- Intro. to Computer Science (CS-101), IITB, *Instructor: Sharat Chandran* Spring 2017

GUEST LECTURES

- Computer Vision (16-720), CMU, *Backpropagation and Optimizers* Fall 2022